

RideWire Job Aid — P0562

Battery Voltage Low (ECM), also referred to in dealer documentation as "ECM voltage low" or "System Voltage Low".

*This job aid is AI-assisted research support compiled from real Harley-Davidson service bulletins, diagnostic manuals, and owner/technician forum reports, cross-checked by three independent AI models. It is **not** a certified repair procedure and does not replace the vehicle's factory service manual. Always verify against manufacturer documentation before performing repairs.*

Platform

Harley-Davidson Twin Cam / Milwaukee-Eight EFI and Carbureted models (ECM), e.g. Dyna, Softail, Touring, Sportster

Code Description¹

Battery Voltage Low (ECM), also referred to in dealer documentation as "ECM voltage low" or "System Voltage Low". Per the Harley-Davidson Electrical Diagnostics Manual, DTC P0562 sets when battery voltage is less than 12.2 volts at idle and voltage does not increase when engine speed is greater than 2000 RPM. When P0562 or its high-voltage counterpart P0563 sets, the battery icon in the speedometer illuminates.

Likely Causes (verified sources)

- Charging system malfunction (failed stator or bad voltage regulator/rectifier) preventing voltage from rising above idle level
- Loose, corroded, or high-resistance battery/ground connections
- Weak or failing battery, or excessive battery draw / extended idling in heavy traffic causing voltage sag
- TSSM/HFSM (security module) fault, which per the HD manual can also set P0562 or P0563

Recommended First Check²

Per Harley-Davidson's official Low/No Charging Diagnostics bulletin (M1507), the first step is to perform a battery test (load/charge test); if the battery passes, proceed to inspect and clean the main DC and AC charging-system connectors, then measure battery voltage with the engine running at 3,000 RPM to confirm it exceeds 13.8V before suspecting the stator or voltage regulator.

Source confidence: Strong: verified against the actual quoted text of the 2008 Dyna Model Electrical Diagnostics Manual (via [hdforums.com](https://www.hdforums.com), a well-established Harley owner/technical forum whose thread reproduces the manual's exact detecting-condition and diagnostic-tip language), an official Harley-Davidson technical service bulletin (M1507, hosted by NHTSA) covering the same charging-system fault family on 2017-2023 Touring/Trike and 2018-2023 Softail platforms, and a second independent forum thread reproducing the identical manual language.

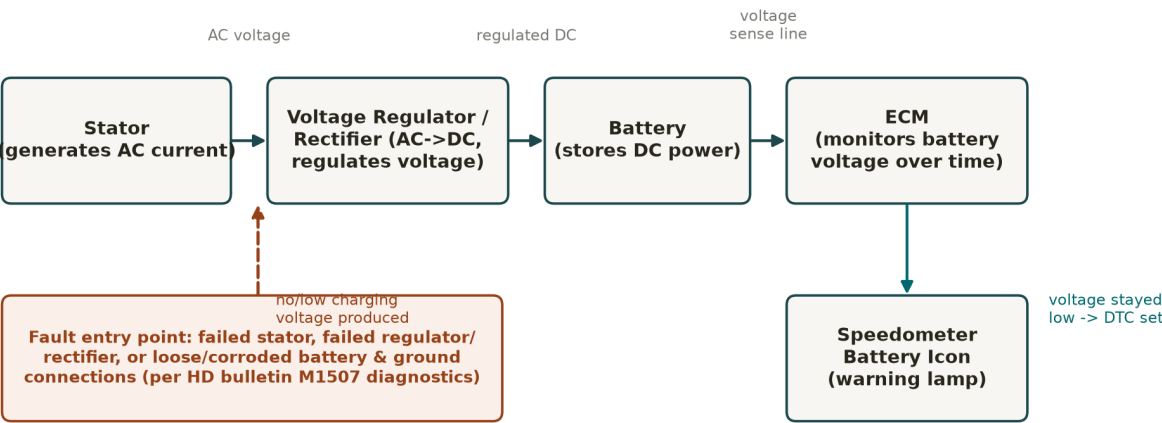
Reference Schematic

Generic principle-level diagram — not a manufacturer-exact wiring diagram for any specific make/model/year. Always verify against the factory service manual.

GENERIC REFERENCE: P0562 Charging System / Battery Voltage Low

Shows the PRINCIPLE of how the charging circuit is monitored - not a vehicle-specific wiring diagram

Result: ECM sees battery voltage stay below ~12.2V at idle and it does not rise above ~2000 RPM -> sets P0562 'Battery Voltage Low' and lights the speedometer battery icon



Reference only - always verify against the actual vehicle's factory service manual before repair.

Multi-AI Consensus (3 independent models)

“Agreement” below is a plain keyword-overlap check on each model's own wording for this single query — it is NOT a validated machine-learning accuracy score and has not been checked against confirmed repair outcomes. Read each model's own words in the table for the actual reasoning.

Model	Primary Root Cause	Urgency	Recommended First Check
Claude Sonnet 5	Weak, undercharged, or failing battery unable to hold voltage above 12.2V at idle	medium	Load-test the battery and measure static/cranking voltage; if battery is good, check charging system output (stator AC voltage and regulated DC voltage at battery) at idle and at 2000+ RPM
GPT-5.6 Terra	Charging system output failure (voltage regulator or stator)	high	Measure voltage directly across the battery terminals at idle and again above 2000 RPM with a digital multimeter; it should rise when engine speed increases.
Gemini 3.1 Pro	Failed voltage regulator or stator	high	Test battery voltage with a multimeter across the terminals at rest and at 2000+ RPM to confirm lack of charging.

Root cause agreement: 3 of 3 models shared the keyword(s) [‘voltage’] in their primary root cause

Urgency values reported: medium, high, high (models differ)

Sources

- 1. <https://www.hdforums.com/forum/dyna-glide-models/279560-need-help-with-code-p0562.html>

2. <https://static.nhtsa.gov/odi/tsbs/2022/MC-10228742-0001.pdf>

3. <https://www.hdforums.com/forum/sportster-models/1202901-2011-sportster-p0562-code.html>

Superscript 1-2 in the text above refer to the same verified source set backing the code description and first-check procedure, drawn from the sources listed here.